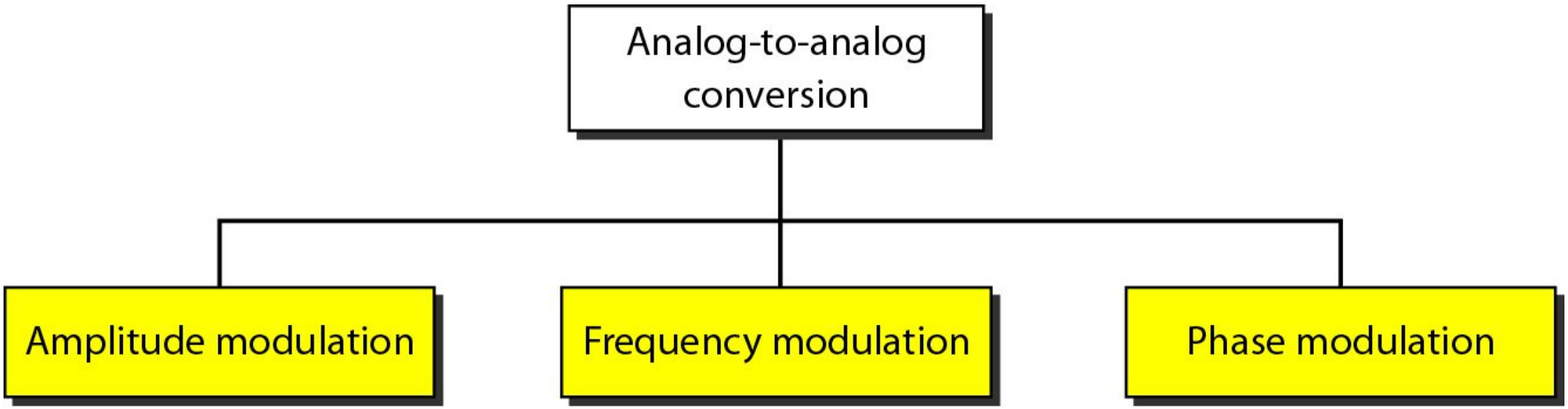
Data Communication
CSE 2211

Chapter 5 : Analog Transmission (Analog to Analog)

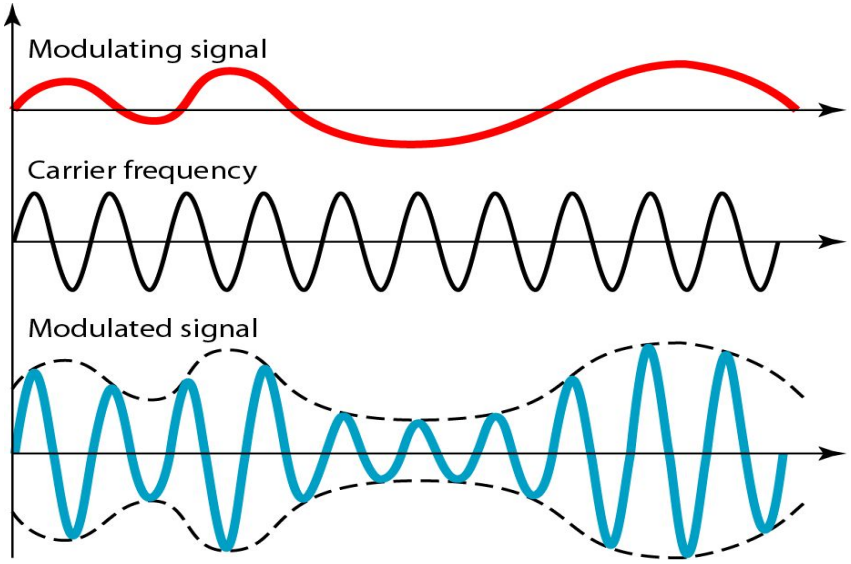
5-2 ANALOG to ANALOG Conversion

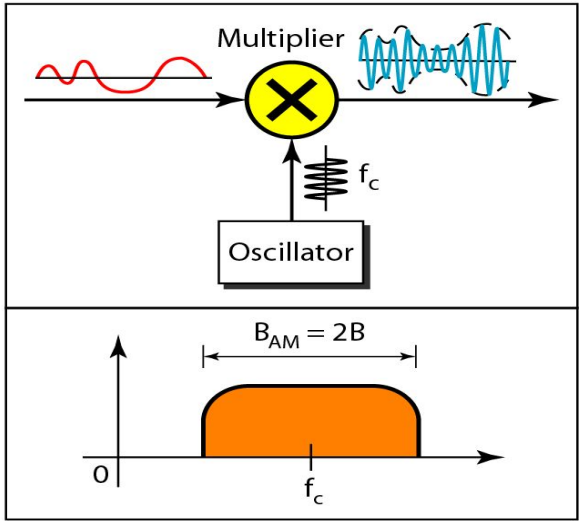
- *Analog-to-analog conversion is the representation of analog information by an analog signal.*
- *Analog to analog modulation is needed if the medium is bandpass in nature or if only a bandpass channel is available to us.*



Amplitude Modulaton

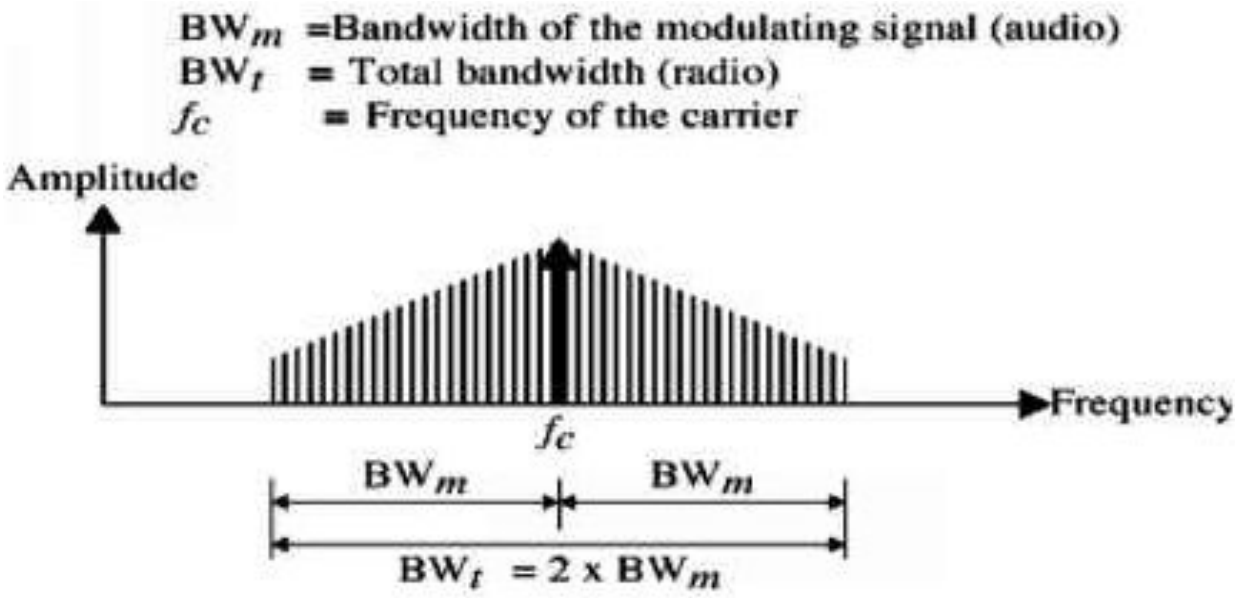
- *In AM transmission, the amplitude of carrier signal varies with the changing amplitudes of the modulating signal.*
- *The frequency and phase of the carrier remain the same.*
- *Normally implemented by a simple multiplier.*
- *Only the amplitude changes to follow variations in the information.*
- *The modulating signal becomes an envelope to the carrier.*





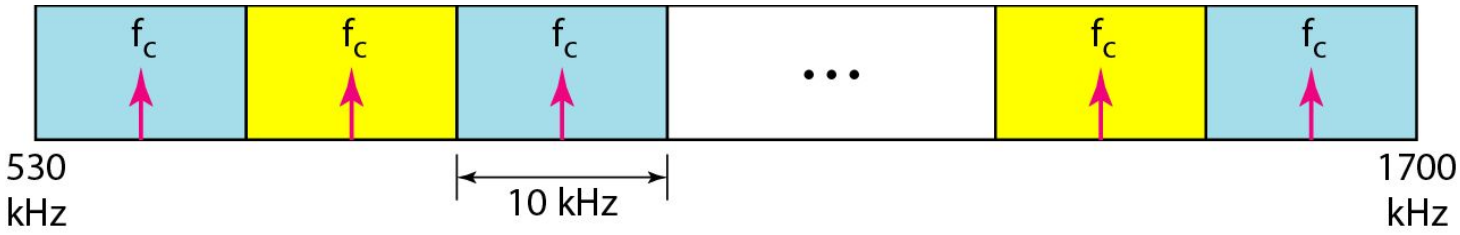
Bandwidth of AM

The bandwidth of an AM signal is equal to twice the bandwidth of the modulating signal and covers a range centered around the carrier frequency.



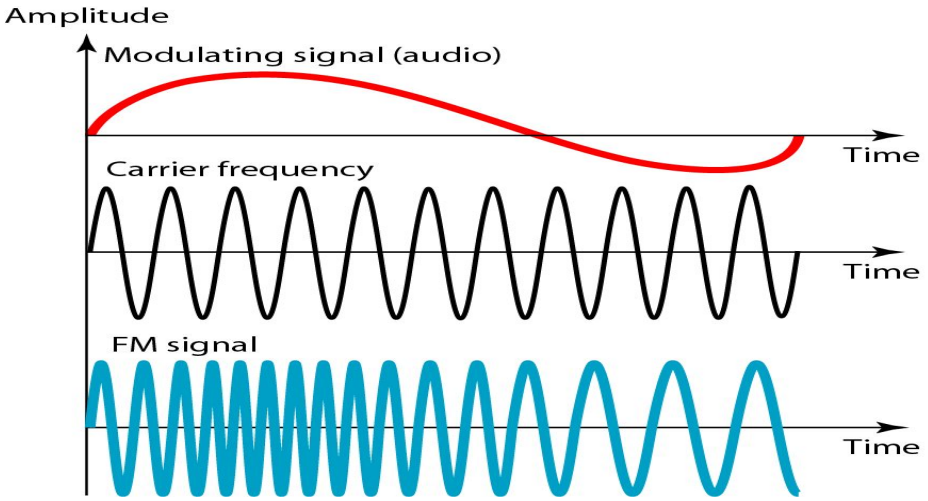
AM Band Allocation

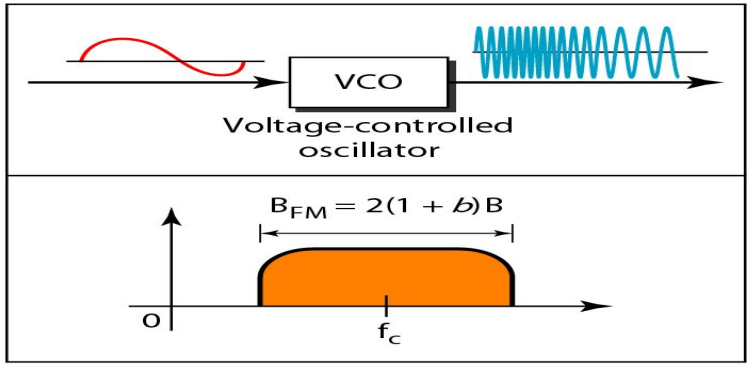
- The bandwidth of an audio signal is usually 5 kHz.
- An AM radio station needs a bandwidth of 2 x 5 kHz = 10 kHz.
- FCC allows 10 kHz for each AM station.
- AM carrier frequency: 530 ~ 1700 kHz (1.7 MHz). How many AM station??
- Each station's carrier frequency must be separated by at least 10 kHz (one AM bandwidth) to avoid interference.
- If one station uses a carrier frequency of 1100 kHz, the next station's carrier frequency cannot be lower than 1110 kHz.



Frequency Modulation

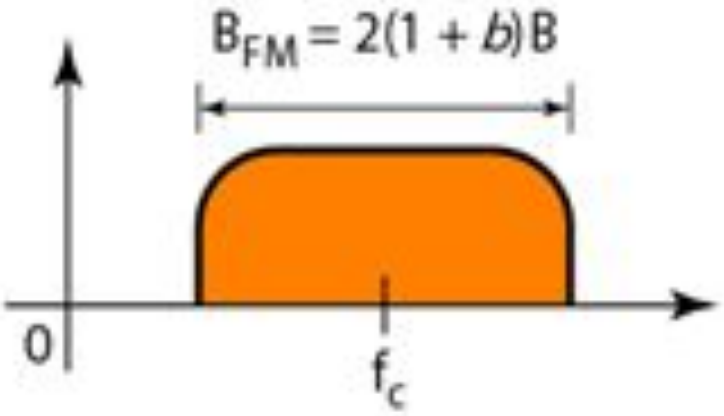
- The frequency of the carrier signal varies with the changing amplitude of the modulating signal.
- Amplitude and phase of the carrier signal remain constant.
- Normally implemented by a voltage-controlled oscillator as with FSK.
- Frequency of the oscillator changes according to the input voltage (amplitude) of the modulating signal.





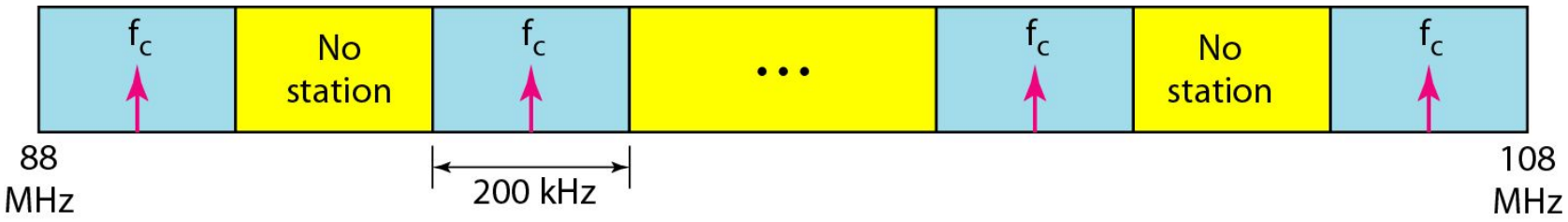
Bandwidth of FM

- Actual bandwidth is difficult to determine exactly.
- The total bandwidth required for FM can be determined empirically as $B_{FM} = 2(1 + \beta)B$.
- β is a factor depends on modulation technique with a common value of 4.
- The bandwidth of an FM signal, is approximately 10 times the bandwidth of the modulating signal.



FM Bandwidth allocation

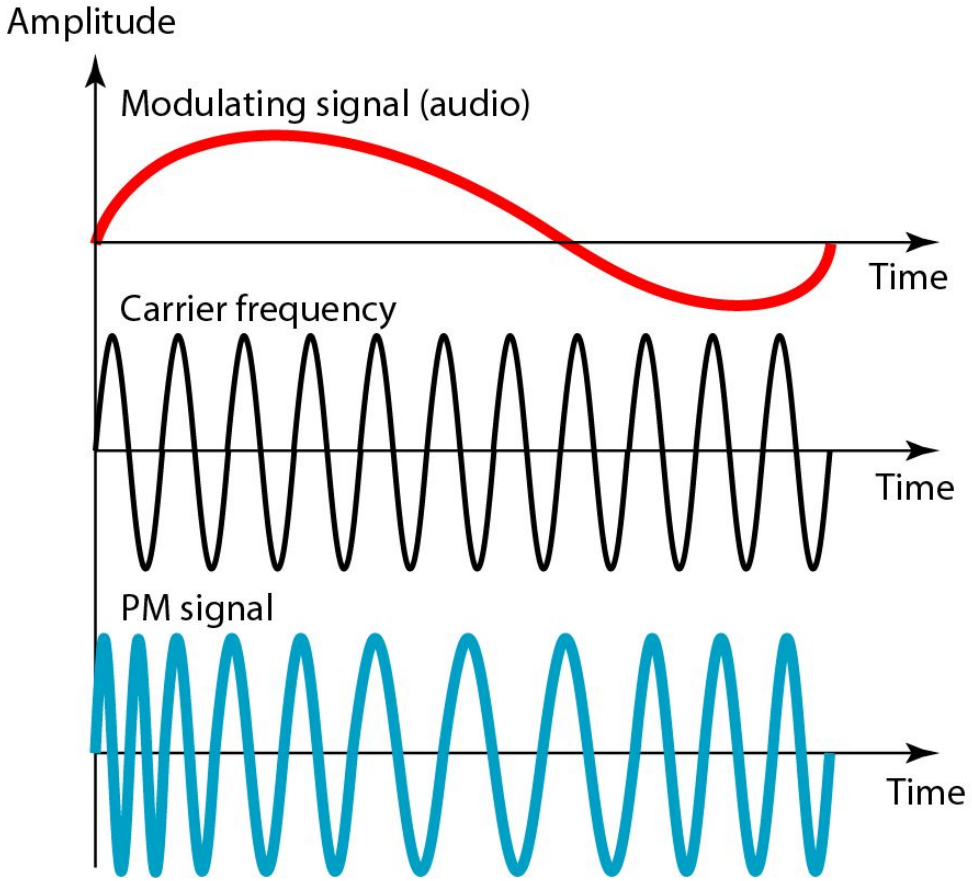
- The bandwidth of a stereo audio (speech and music) broadcast is almost $B = 15$ kHz.
- FCC allows 200 kHz for each station.
  - $BFM = 2(1 + \beta)B$, $\beta = 4$ with some extra guard band.
- FM carrier frequency ranges from 88 to108 MHz.
  - Separated by at least 200 kHz without overlapping.
- To create more privacy, FCC requires that in a given area, only alternate bandwidth allocations may be used.
  - The others remain unused to prevent any possibility of two stations interfering with each other.
- In spectrum of 88 ~108 MHz, 100potential FM bandwidths in an area. Only 50can operate at anyone time.

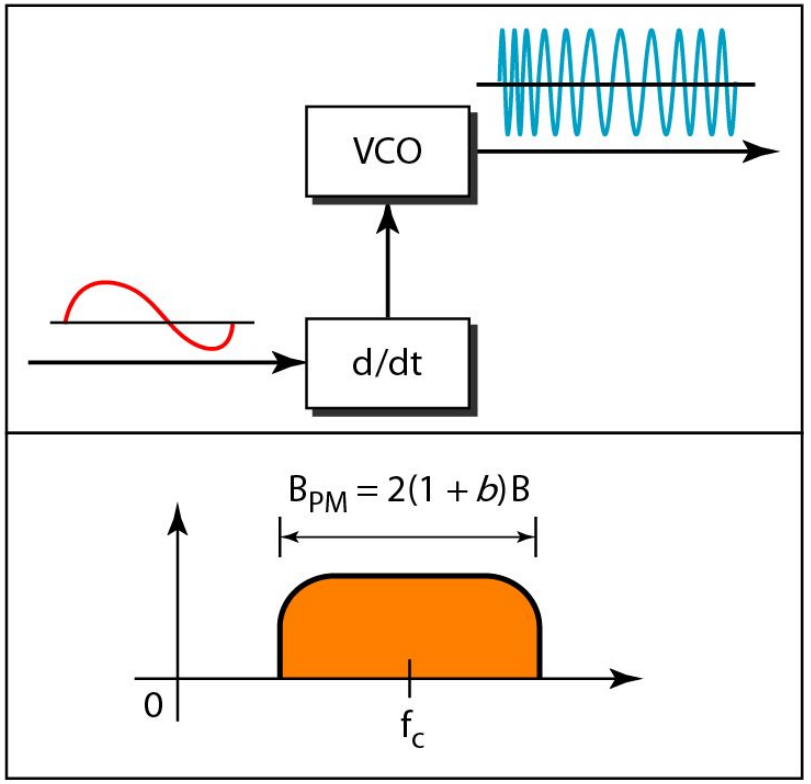


Phase Modulation (PM)

- The phase of the carrier signal varies with the changing amplitude of the modulating signal.
- Amplitude and frequency of the carrier signal remain constant.
- It can proved mathematically that PM is the same as FM with one difference.
- In FM, the instantaneous change in the carrier frequency is proportional to the amplitude of the modulating signal.
- In PM, the instantaneous change in the carrier frequency is proportional to the derivative of the amplitude of the modulating signal.

Figure 5.20 *Phase modulation*





Note

The total bandwidth required for PM can be determined from the bandwidth and maximum amplitude of the modulating signal:

$$B_{PM} = 2(1 + \beta)B.$$

Where $\beta = 2$ most often.